

Thm. Riemann's Theorem.

Let  $f$  be a complex function having an isolated singularity at  $z=z_0$ .

If  $f$  is bounded for some deleted neighborhood of  $z_0$ ,

then  $z_0$  is removable.

Proof. By the hypothesis, for some  $R > 0$ ,  $f$  is analytic and bounded in

$$0 < |z - z_0| < R.$$

Given  $z_1 \in B_R(z_0) \setminus \{z_0\}$ , let  $r > 0$  s.t.  $r < |z_1 - z_0| < R$ . So,  $f$  is analytic and bounded in the annulus between  $C_1: |z - z_0| = R$  and  $C_r: |z - z_0| = r$ .

By the Cauchy's integral formula,

$$f(z_1) = \frac{1}{2\pi i} \cdot \left[ \int_{C_1} \frac{f(u)}{u - z_1} du - \int_{C_r} \frac{f(u)}{u - z_1} du \right]$$

Meanwhile, by the ML inequality,

$$\left| \int_{C_r} \frac{f(u)}{u - z_1} du \right| \leq \sup |f| \cdot 2\pi r \cdot \frac{1}{|z_1 - z_0| - r} \rightarrow 0 \text{ as } r \rightarrow 0.$$

$$\text{So, } f(z_1) = \frac{1}{2\pi i} \int_{C_1} \frac{f(u)}{u - z_1} du.$$

Since  $z_1$  was arbitrary, we can denote that

$$f(z) = \frac{1}{2\pi i} \cdot \int_{C_1} \frac{f(u)}{u - z} du \quad \text{for all } 0 < |z - z_0| < R,$$

$$\text{So, define } f(z_0) = \frac{1}{2\pi i} \cdot \int_{C_1} \frac{f(u)}{u - z_0} du, \quad \text{we obtain the result.}$$

[Thm. Suppose that  $f$  has an isolated singularity at  $z_0$ . TFAE:

1) For some  $m \in \mathbb{N}$ ,  $\lim_{z \rightarrow z_0} (z - z_0)^m \cdot f(z) = A \in \mathbb{C} \setminus \{0\}$ . (That is,  $z_0$  is pole of order  $m$ ).

2).  $f(z) \rightarrow \infty$  as  $z \rightarrow z_0$

3).  $f(z) = g(z)/(z - z_0)^m$  for some analytic  $g(z)$  and  $m \in \mathbb{N}$ , with  $g(z_0) \neq 0$ .

Proof. Suppose that there exists  $m \in \mathbb{N}$  s.t.  $\lim_{z \rightarrow z_0} (z - z_0)^m f(z)$  exists with non-zero.

Given  $\varepsilon = |A|/2$ , there exists  $\delta > 0$  such that

$$|z - z_0| < \delta \Rightarrow |(z - z_0)^m \cdot f(z) - A| < |A|$$

So, using the triangle inequality, for  $0 < |z - z_0| < \delta$ ,

$$|A| - |z - z_0|^m \cdot |f(z)| < \frac{|A|}{2}$$

$$\Rightarrow \frac{|A|}{2\delta^m} < \frac{|A|}{2|z - z_0|^m} < |f(z)|$$

Thus, taking  $\delta \rightarrow 0$ , then  $f(z) \rightarrow \infty$ . [1)  $\Rightarrow$  2).

Suppose that  $f(z) \rightarrow \infty$  as  $z \rightarrow z_0$ . Put  $g(z) = 1/f(z)$  in  $B_\delta(z_0) \setminus \{z_0\}$  for some  $\delta > 0$ , which satisfies  $f(z) \neq 0$ . (If no such neighbor exists, then  $f$  is identically zero.)

Now, since  $f(z) \rightarrow \infty$  as  $z \rightarrow z_0$ , so  $g(z) \rightarrow 0$  as  $z \rightarrow z_0$ , so  $z_0$  is removable to  $g$ .

Then, the Taylor expansion

$$g(z) = a_0 + a_1(z - z_0) + \dots \quad (|z - z_0| < \delta).$$

is valid. Since  $g(z) \neq 0$  in  $B_\delta(z_0) \setminus \{z_0\}$ , so not all the coefficients are zero.

Let  $m$  be the smallest integer s.t.  $a_m \neq 0$ . ( $m \geq 1$  because  $\lim_{z \rightarrow z_0} g(z) = a_0 = 0$ ).

So that  $g(z) = a_m(z - z_0)^m + a_{m+1}(z - z_0)^{m+1} + \dots$

Now,  $\frac{1}{(z - z_0)^m f(z)} = a_m + a_{m+1}(z - z_0) + \dots \rightarrow a_m$  as  $z \rightarrow z_0$ ,

that is,  $(z - z_0)^m f(z) \rightarrow a_m^{-1}$  as  $z \rightarrow z_0$ . [2)  $\Rightarrow$  3).

3)  $\Rightarrow$  2) is directly, the analytic  $g$  is continuous at  $z_0$ .

Isolated singularity at  $\infty$ .

If  $f$  is analytic for all  $R < |z| < \infty$ , for some  $R > 0$ , then we say that  $f$  has an isolated singularity at  $\infty$ .

For this  $R > 0$ , observe that

$f(z)$  is analytic at  $|z| > R$  if and only if  $f(\frac{1}{z})$  is analytic at  $|z| < \frac{1}{R}$ .

$f(z)$  has an isolated singularity at  $\infty$  iff  $f(\frac{1}{z})$  has an isolated singularity at 0.

Thm. Let  $f$  be a non-constant entire function. Then,

$f$  is polynomial if and only if  $f(z) \rightarrow \infty$  as  $z \rightarrow \infty$ .

Proof. Suppose that  $f$  is a polynomial. WLOG, let  $f(z) = a_0 + a_1 z + \dots + a_n z^n$ . ( $a_n \neq 0$ ).

Then,  $f(\frac{1}{z}) = \frac{a_n}{z^n} + \frac{a_{n-1}}{z^{n-1}} + \dots + \frac{a_1}{z} + a_0$  so that  $\lim_{z \rightarrow 0} z^n \cdot f(\frac{1}{z}) = a_n \neq 0$ .

So,  $f(z)$  has a pole at  $\infty$ , so  $\lim_{z \rightarrow \infty} f(z) = \infty$ .

Conversely, suppose that:  $f(\frac{1}{z})$  has a pole of order  $m$  at 0. That is,

$$\lim_{z \rightarrow 0} \frac{f(z)}{z^m} = A \in \mathbb{C} \setminus \{0\}.$$

That is, for some  $R > 0$ ,  $|z| \geq R \Rightarrow \left| \frac{f(z)}{z^m} - A \right| < 1$

$$\Rightarrow \left| \frac{f(z)}{z^m} \right| < 1 + |A|$$

$$\Rightarrow |f(z)| < (1 + |A|) \cdot z^m$$

So,  $f$  is a polynomial of degree at most  $m$ .

# Singularities of rational functions.

Let  $P$  and  $Q$  be polynomials over  $\mathbb{C}$ ,  $P(z) = a_0 + a_1 z + \dots + a_n z^n$ ,  $Q(z) = b_0 + \dots + b_m z^m$ ,  
and  $f(z) = P(z)/Q(z)$ . Then,  $(a_n, b_m \neq 0)$ .

- $f(z)$  has a removable singularity at  $\infty$
- iff  $f(1/z)$  has a removable singularity at 0
- iff for some  $\epsilon > 0$  and  $M > 0$ ,  $0 < |1/z| < \epsilon \Rightarrow |f(1/z)| \leq M$ .
- iff  $|z| > \frac{1}{\epsilon} \Rightarrow |f(z)| \leq M$ .
- iff  $\deg P(z) \leq \deg Q(z)$ .

- $f(z)$  has a pole of order  $k$  at  $\infty$
- iff  $f(1/z)$  has a pole of order  $k$  at 0.
- iff  $f(1/z) = \frac{P(1/z)}{Q(1/z)} = \sum_{i=-k}^{\infty} C_i \cdot z^i$  for some  $\{C_i\}_{i=-k}^{\infty}$

iff  $P(1/z) = \sum_{i=0}^n a_i \cdot \frac{1}{z^i} = \left( \sum_{i=0}^m b_i \cdot \frac{1}{z^i} \right) \cdot \left( \sum_{i=-k}^{\infty} C_i z^i \right)$

- iff  $n = m + k$ , that is, the order  $k = \deg P - \deg Q > 0$ .

Long division.

Consider a function represented by

$$f(z) = \left( \sum_{n=0}^{\infty} a_n z^n \right) / \left( \sum_{n=0}^{\infty} b_n z^n \right) = \frac{a_0 + a_1 z + a_2 z^2 + \dots}{b_0 + b_1 z + b_2 z^2 + \dots} \quad (b_0 \neq 0)$$

Since  $b_0 \neq 0$ , this  $f$  is analytic at  $z=0$ , so  $f$  can be written as

$$f(z) = \frac{a_0 + a_1 z + \dots}{b_0 + b_1 z + \dots} = c_0 + c_1 z + \dots$$

so that

$$\left( \sum_{n=0}^{\infty} a_n z^n \right) = \left( \sum_{n=0}^{\infty} b_n z^n \right) \cdot \left( \sum_{n=0}^{\infty} c_n z^n \right), \text{ in some neighbor of the origin, thus}$$

the uniqueness theorem gives

$$a_n = \sum_{k=0}^n b_k \cdot c_{n-k}, \text{ the Cauchy product.}$$

Example.

Let  $f(z) = \frac{\pi \cdot \cot \pi z}{z^4}$ . Then,

$$\begin{aligned} f(z) &= \frac{\pi}{z^4} \cdot \left( \frac{1 - \frac{\pi^2}{2!} z^2 + \frac{\pi^4}{4!} z^4 + \dots}{\pi z - \frac{\pi^3}{3!} z^3 - \frac{\pi^5}{5!} z^5 + \dots} \right) \\ &= \frac{1}{z^5} \cdot \left( \frac{1 - \frac{\pi^2}{2} z^2 + \frac{\pi^4}{4!} z^4 - \dots}{1 - \frac{\pi^2}{3!} z^2 + \frac{\pi^4}{5!} z^4 - \dots} \right) \end{aligned}$$

$$1 = 1 \cdot c_0 \Rightarrow c_0 = 1$$

$$a_1 = b_0 c_1 + b_1 c_0 = c_1 + 0 = 0. \Rightarrow c_1 = 0.$$

$$a_2 = b_0 c_2 + b_1 c_1 + b_2 c_0 = c_2 + 0 + -\frac{\pi^2}{8} \cdot 1 = -\frac{\pi^2}{8} \Rightarrow c_2 = -\frac{\pi^2}{8}.$$

$$a_3 = b_0 c_3 + b_1 c_2 + b_2 c_1 + b_3 c_0 = c_3 + 0 + -\frac{\pi^2}{8} \cdot 0 + 0 = 0 \Rightarrow c_3 = 0.$$

$$a_4 = b_0 c_4 + b_1 c_3 + b_2 c_2 + b_3 c_1 + b_4 c_0 = c_4 + c_2 \cdot -\frac{\pi^2}{8} + \frac{\pi^4}{5!} = c_4 + \pi^4 \cdot (1/18 + 1/120) = \frac{\pi^4}{24}.$$

$$\Rightarrow c_4 = \pi^4 \left( \frac{1}{24} - \frac{1}{18} - \frac{1}{120} \right) = \pi^4 \cdot \left( -\frac{1}{45} \right) = -\frac{1}{45} \pi^4.$$

Consequently, the principal part of  $f$  is

$$\frac{1}{z^5} - \frac{\pi^2}{3} \cdot \frac{1}{z^3} - \frac{\pi^4}{45} \cdot \frac{1}{z}.$$